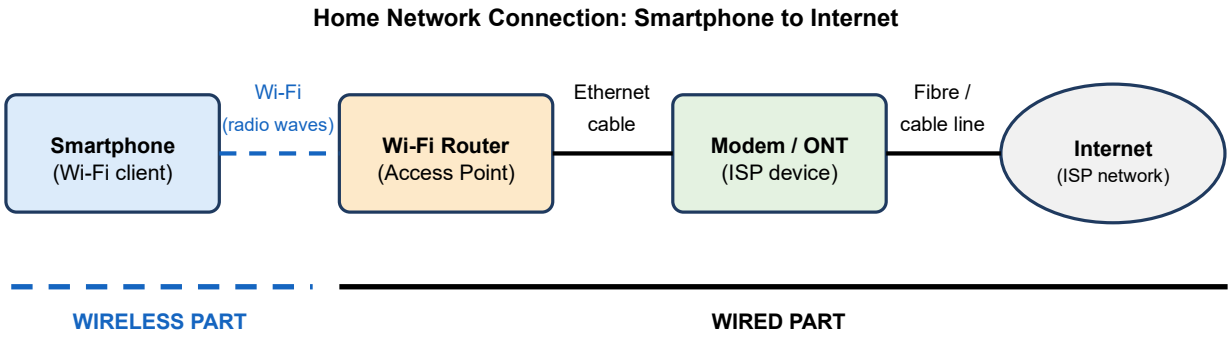
WIRELESS NETWORKING

TASK-1 List 3 popular Wi-Fi standards (e.g., 802.11n, 802.11ac, 802.11ax) and for each, write one key difference in speed or frequency band.

Standard	Wi-Fi name	Frequency band	Max theoretical speed	One key difference
802.11n	Wi-Fi 4	2.4 GHz and 5 GHz	Up to 600 Mbps	First standard to use MIMO (multiple antennas) and to allow dual-band operation, giving a big speed jump over 802.11g.
802.11ac	Wi-Fi 5	5 GHz only	Up to about 3.5 Gbps	Works only on the 5 GHz band with wider channels (80/160 MHz), so it is faster but has shorter range through walls than 2.4 GHz.
802.11ax	Wi-Fi 6 / 6E	2.4 GHz, 5 GHz (and 6 GHz in Wi-Fi 6E)	Up to 9.6 Gbps	Uses OFDMA, so one router can serve many devices at the same time efficiently; Wi-Fi 6E adds the new 6 GHz band.

WIRELESS NETWORKING

TASK-2 Draw a simple diagram (on paper or using any drawing tool) showing how your smartphone connects to the internet via a Wi-Fi router at home, labeling the wireless and wired parts of the network.



Explanation

- **Wireless part:** the smartphone connects to the Wi-Fi router over radio waves (2.4 GHz / 5 GHz). No cable is used here.
- **Wired part:** the router is connected by an Ethernet cable to the modem/ONT, which connects through the ISP's fibre or cable line to the Internet.

WIRELESS NETWORKING

TASK-3 Scan for available Wi-Fi networks using your laptop or phone, note down the SSID, signal strength, and security type (WPA2, WPA3, etc.) for at least 3 networks, and explain which one you would choose and why.Hint Consider both signal strength and security when choosing.

No.	SSID (network name)	Signal strength	Security type	Band
1	Home_Fiber_5G	-42 dBm (excellent)	WPA3-Personal	5 GHz
2	Neighbour_WiFi	-68 dBm (fair)	WPA2-Personal	2.4 GHz
3	Cafe_Guest	-55 dBm (good)	Open (no security)	2.4 GHz

Signal strength is measured in dBm. A value closer to 0 is stronger (about -30 is excellent, -67 is good enough for video streaming, -80 or lower is weak).

Network I would choose: Home_Fiber_5G

- **Signal:** -42 dBm is the strongest of the three, so speed will be high and stable.
- **Security:** WPA3 is the newest and strongest standard, with better protection against password-guessing attacks.

- **Why not the others:** Cafe_Guest is open, so anyone can capture the traffic sent over it. Neighbour_WiFi uses WPA2, which is good, but its signal is weaker and it is not my network.

TASK-4 Explain in your own words what 'SSID hiding' is in Wi-Fi networks and discuss one advantage and one disadvantage of hiding your Wi-Fi SSID.

What is SSID hiding?

Every Wi-Fi router regularly broadcasts its network name (SSID) in small packets called beacon frames, so nearby devices can list it. SSID hiding (also called disabling SSID broadcast) tells the router to stop announcing the name. The network does not appear in the normal list of available Wi-Fi networks, and a user must type the exact SSID and password manually to connect.

	Explanation
One advantage	The network is not visible to casual users or neighbours scanning for Wi-Fi, so it looks less attractive and gets fewer random connection attempts.
One disadvantage	It is not real security. The SSID is still sent inside other packets when a device connects, so free tools such as Wireshark can reveal it in seconds. Also, hidden-network devices keep sending probe requests with the name, and users find it harder to connect.